FRANCISCO PEREIRA

Software Developer

Python | LLM applications

Portugal

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OVERVIEW

Software developer maintaining and modernising legacy core products in Python, including LLM-backed conversational features. In a previous R&D role I built the company's first AI prototypes: a retrieval-augmented assistant over the internal Confluence knowledge base and a self-hosted LLM inference API running on local hardware.
Peer-reviewed publication in mixed reality for smart manufacturing. Starting a postgraduate degree in Applied Machine Learning in September 2026, working towards a Machine Learning Engineering role.

EXPERIENCE

Jan 2026-Present **Software Developer** — **GoTVee | Remote**
- Maintain and modernise legacy core products, refactoring and sustaining systems the business depends on.
- Build and maintain features in Python, including LLM-backed conversational features.
- Maintain and extend a multi-tenant enterprise video platform serving thousands of videos for the company's largest client: bug resolution, new features and ongoing code maintenance.

Jan 2025-Dec 2025 **Research and Development Technician** — **GoTVee | Bragança, Hybrid**
- Built an internal retrieval-augmented (RAG) assistant over the company's Confluence knowledge base, indexing 30+ spaces for natural-language search.
- Designed and deployed a self-hosted LLM inference API on an RTX 3090 (24 GB), using Ollama and LangChain with vector search over locally generated embeddings, removing any dependency on external model providers.
- Prototyped a per-video chat widget for the enterprise video platform, delivered to staging as a working MVP.

Jan 2024-Dec 2024 **Software Developer Internship** — **GoTVee | Bragança, Hybrid**
Developed proof-of-concepts (PoCs) and minimum viable products (MVPs) for AI systems using large language models, building both the Python services and the interfaces used to evaluate them.

Jul 2023-Sep 2023 **Research Fellowship at CeDRI** — **CeDRI, Instituto Politécnico de Bragança**
Contributed to a C# and OpenGL tool in Unity 3D for the OpenZDM project, improving defect detection in automotive assembly and body shop quality assurance.

Jul 2022-Jan 2023 **Research Fellowship at CeDRI** — **CeDRI, Instituto Politécnico de Bragança**
Developed mixed reality (AR and VR) applications in Unity 3D.

EDUCATION

Sep 2026-2027 **Postgraduate Degree in Applied Machine Learning** — **ISCTE – Instituto Universitário de Lisboa**

Oct 2019-Jul 2023 **Degree in Informatics Engineering** — **Instituto Politécnico de Bragança**
- Scientific Areas: Computer Science, Computer Engineering, Mathematics, Information Systems.
- Honors: Awarded two research fellowships

LANGUAGES

Portuguese - Native
English - Proficient

TECHNICAL SKILLS

Programming: Python, C#, JavaScript.
AI / ML: Self-hosted LLM inference (Ollama), retrieval-augmented generation (RAG) with LangChain, vector embeddings and similarity search (FAISS), CUDA-accelerated inference.
Backend: FastAPI, Django, REST APIs, NGINX.
Data: SQL, NoSQL, vector databases.
Platforms and Tools: Linux, Bash, Docker, CI/CD pipelines, Git (GitHub, GitLab, BitBucket), Google Cloud Platform (GCP), Unity 3D.

PUBLICATIONS

An Augmented Reality Intelligent Guide for the Automotive Industry
Published in Learning Factories of the Future (CFL 2024) | University of Twente, Enschede, The Netherlands (Jul 2024)
- Developed a mixed reality HoloLens app using AI and computer vision to identify and display vehicle inspection zones on a production line.
- Integrated gesture-based interaction and a RESTful API for real-time database updates, enhancing efficiency and safety in smart manufacturing.
DOI: 10.1007/978-3-031-65400-8_23